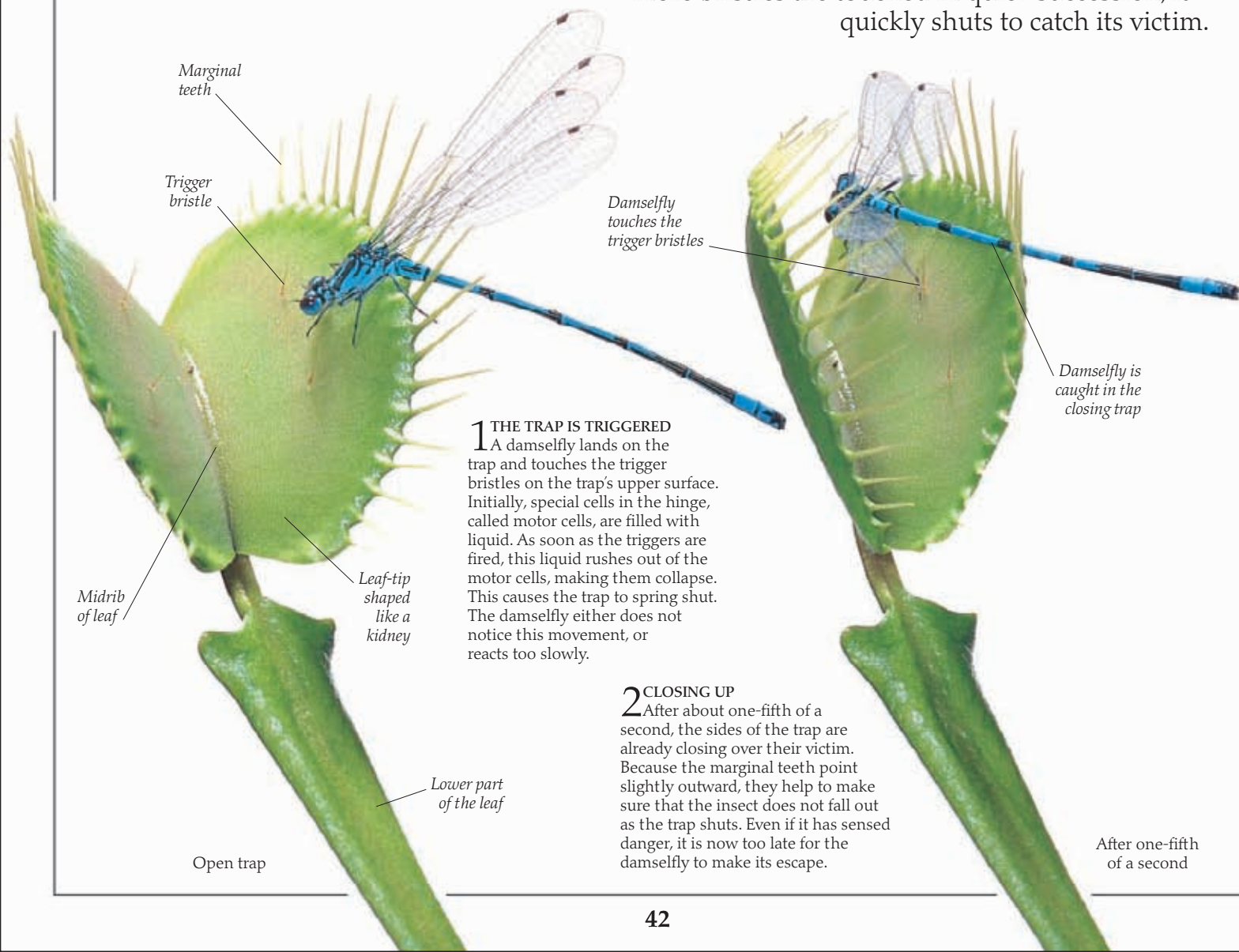




A MOST HORRIBLE TORTURE
Like the Venus flytrap, humans have also been known to inflict a frightful and lingering death on their victims, as this rather gruesome engraving shows.

Caught in a trap

TO AN UNWARY INSECT, the unusually shaped tip of a Venus flytrap leaf appears most inviting. Not only is the insect attracted by what looks like a safe landing place, but it is also tempted by the promise of food in the form of nectar. But it is all a trick. As soon as the insect settles, the leaf-tip springs to life with lightning speed. Within a second, the hapless insect finds itself trapped, as the two halves of the leaf-tip snap shut. There is a second, slower phase of closure after the plant has tested what it has caught using sensory glands on the upper surface of the leaf-tip. If the prey contains protein, the trap closes fully, and digestion begins. The traps of the Venus flytrap are formed by two kidney-shaped lobes at the tip of the leaf, with a hinge formed by the midrib. The whole of the leaf is green and therefore able to photosynthesize (pp. 14–15). Large bristles on the upper surface of the trap work like triggers with a clever device. If just one bristle is touched, by a raindrop for example, the trap stays open. But if two or more bristles are touched in quick succession, it quickly shuts to catch its victim.



1 THE TRAP IS TRIGGERED
A damselfly lands on the trap and touches the trigger bristles on the trap's upper surface. Initially, special cells in the hinge, called motor cells, are filled with liquid. As soon as the triggers are fired, this liquid rushes out of the motor cells, making them collapse. This causes the trap to spring shut. The damselfly either does not notice this movement, or reacts too slowly.

2 CLOSING UP
After about one-fifth of a second, the sides of the trap are already closing over their victim. Because the marginal teeth point slightly outward, they help to make sure that the insect does not fall out as the trap shuts. Even if it has sensed danger, it is now too late for the damselfly to make its escape.

After one-fifth of a second